Speaker script

Broadband Expansion and Small-Firm Productivity: Evidence from a Staggered Municipal Rollout

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Opening 0.5 min

Good morning. I am Astrid Lindqvist, and this is joint work with Tomas Berg. The setting and data are synthetic, but the economic problem is real: distance can keep a small firm from using outside expertise. We ask whether productivity is higher after broadband reaches a municipality, and whether the difference is greatest for firms that depend heavily on communication. This connects digital infrastructure to the way firms organize production.

Live cue: Greeting; coauthor; quiet demo disclaimer; exact empirical questions.

Key question 2.2 min

Think about a small manufacturer that needs an accountant, software support, and help coordinating with suppliers. Hiring each specialist full time makes little sense. A large firm can spread that fixed cost over many workers; a small firm usually has to buy the service outside. The problem is that working with a distant provider takes communication, and communication used to be expensive and slow.

Broadband can change that calculation. Its value is not simply that email arrives faster. It can widen the set of specialists a firm can reach and make day-to-day coordination with a remote provider practical. That may move some tasks across the boundary of the firm. If a better specialist produces more value without a proportional increase in labor, we should see higher value added per worker.

Small firms are a particularly interesting test. They have the strongest reason to look outside for expertise because an internal specialist is costly. At the same time, they may have less money and managerial capacity to reorganize around a new technology. So the sign and size of the productivity response are not obvious in advance.

We ask two linked questions. First, is labor productivity higher after commercial backbone service reaches the firm's municipality? Second, is that difference larger for firms that began with more communication-intensive work? The second question is what gives the first one economic content. If remote coordination is the relevant friction, the firms with more tasks that can use it should display the larger difference.

We observe the common date when service becomes available in each municipality and follow firms around that date. We do not observe each subscription decision. So throughout the talk, "broadband arrival" means local availability. That is the policy change we can measure consistently for every firm.

Live cue: Organization problem; why small firms; two questions; availability.

This paper 1.8 min

Here is what we do. We combine the rollout dates for 284 municipalities with annual records for 9,400 small firms from 2008 through 2019. That gives us a long panel in which the same firm can be followed before and after service reaches its municipality.

The first result compares each early rollout cohort only with places that were still waiting. The answer is the same in every wave: each of the first five rollout cohorts is more productive after the backbone arrives than municipalities connected at least three years later. Firms are producing more value from the same labor input. A pooled two-way regression summarizes the whole rollout as 0.034 log points, about 3.4 percent. Because arrival is staggered and the differences grow over time, that pooled number sits at the low end of the cohort comparisons.

We then build a simple task-sourcing model to ask which firms should benefit most. Broadband lowers the cost of coordinating with a specialist at a distance. A firm gains more from that change when a larger share of its work can use the specialist. The data line up with that prediction. A one-standard-deviation increase in pre-rollout communication intensity adds 0.0064 log points to the post-arrival difference, about one-fifth of the pooled association.

The contribution is to connect a place-based infrastructure change to a within-firm organizational margin. The cohort comparisons tell us the scale of the productivity pattern. The model and the cross-firm comparison tell us where that pattern is strongest and why. Together they suggest that the productivity differences are largest where communication frictions bind.

Live cue: Data link; cohort result and pooled 3.4 percent; task-sourcing prediction; contribution.

Roadmap 0.3 min

I will make three stops. First, the rollout and the firm data. Second, the timing comparison and the main productivity result. Third, the task-sourcing model and the cross-firm evidence that speaks to it.

Live cue: Setting and data; design and evidence; model and interpretation.

Broadband reached every municipality in eight waves 2.2 min

The rollout gives us a simple piece of timing variation. Calendar year is on the horizontal axis, and the vertical axis is the cumulative share of Nordavia's 284 municipalities with commercial backbone service. Every step is a new wave of municipalities coming online.

Three dates tell most of the story. The first wave arrives in 2011, half the municipalities are connected by 2014, and the last wave arrives in 2018. In the middle of the sample, firms in one part of the country have service while firms elsewhere are still waiting. We use that overlap to compare firms at different points in the rollout. By the end, every municipality is connected, so the comparison comes from earlier and later arrival rather than from a group that never receives service.

The 2010 Digital Infrastructure Act set the goal of universal coverage. The RDDF then recorded when each backbone segment entered commercial service. Construction followed geography—distance and terrain mattered—but also a policy priority score. The score moved low-productivity municipalities with stronger projected growth toward the front of the queue. That detail will be useful when we look at the event-time pattern.

The policy object is local access to the backbone. The paper reports take-up above 90 percent within two years among firms with more than five employees, so service availability often became actual access in that group. But the common, consistently measured change is the municipal service date. We use that date as the starting point and then ask how firm outcomes move around it.

The staircase is therefore more than institutional background. It creates the staggered timing in the empirical design. The next step is to connect those municipal dates to the same firms over time and to measure which firms had the greatest potential to use communication at a distance.

Live cue: Axes and three dates; source of variation; selected timing; availability.

Data 2.4 min

We build the analysis from three administrative sources. The business register tells us which firms exist and where their headquarters are. Audited accounts give us value added and investment. Employer–employee records give us employment and the occupation of each worker. After linking them, one observation is a firm in a year.

The panel follows 9,400 small firms from 2008 through 2019, for 112,800 firm-year observations. Firms start with 3 to 49 employees and have positive sales. The panel is balanced, so every firm appears in all twelve years. This is helpful for a within-firm design: the set of firms does not change as we move from the pre-rollout to the post-rollout period.

Our outcome is real value added per worker. We deflate value added using the producer-price index for the firm's industry, divide by full-time-equivalent employment, and take logs. The object is easy to interpret: how much real value the firm produces for each unit of labor. It captures changes in the productivity of the firm's overall production process, including changes in how inputs and tasks are organized.

We assign each firm the commercial-service date of its headquarters municipality. The broadband indicator turns on in that year and stays on. This gives us a common measure of when the local communication constraint changes, even though individual subscription dates are not recorded.

The final ingredient is communication intensity. We take each firm's 2008 wage-bill share in occupations with above-median communication requirements. Client contact, supplier coordination, and remote support are examples. The measure is fixed in 2008, before any municipality receives service. So a firm that later changes its workforce cannot move itself into a different initial category.

These links give us exactly the three objects the argument needs: an outcome, a service-arrival date, and a pre-rollout measure of the tasks that may benefit from cheaper remote coordination. We now use the timing to compare changes within firms.

Live cue: Linked records; balanced survivor panel; outcome; exposure; 2008 task mix.

The design follows firms as local backbone service arrives 2.5 min

The design asks how a firm's productivity changes when local service arrives. On the left of the equation is log real value added per worker. The colored term switches on in the service year for the firm's municipality. Beta compares the firm's pre- and post-arrival years with other rollout cohorts in the same year.

Firm fixed effects make this a within-firm comparison. They absorb features such as a firm's long-run product mix, management style, and location. Year fixed effects absorb national shocks: a recession, a common change in input prices, or an economy-wide technology shift. We are therefore asking whether the same firm moves differently when its municipality reaches a different point on the rollout clock.

The pooled regression mixes two kinds of comparisons. Some set newly served firms against places still waiting; others use places served earlier as the benchmark. When the difference grows with time since arrival, those second comparisons pull the pooled coefficient down. So our main evidence compares each cohort only with municipalities connected at least three years later.

The key economic assumption is about the path without broadband. Earlier- and later-served municipalities should otherwise have followed similar productivity trends, after we remove firm and year effects. This is why the rollout rule matters. Places with stronger projected growth moved up the queue. If those projections were informative, earlier cohorts might already have been on a different path.

We examine that issue directly with event time. Rather than collapse all post-arrival years into one indicator, the event study shows how productivity evolves in each year before and after service. The pre-arrival coefficients check whether the cohorts were already moving differently as arrival approached; a flat pre-period would not prove the assumption, but a slope is a warning. The post-arrival coefficients show whether the trajectory changes or continues.

Finally, service timing is assigned at the municipality level. Firms in the same municipality can share local shocks, so the most informative standard errors allow their outcomes to move together. The pooled table in the appendix reports that municipality-level clustering. I will use the cohort comparisons for scale and the event study for the shape of the timing pattern.

Live cue: Within-firm timing comparison; fixed effects; path without broadband; clustering.

Each 2011–2015 cohort is more productive after arrival 2.2 min

This figure is the paper's cleanest timing comparison. Each bar is one rollout cohort, from 2011 through 2015, compared only with municipalities that were connected at least three years later. No place that already has service is used as a control, so a growing effect in earlier cohorts cannot contaminate the comparison.

Read the bars from left to right. Every cohort is above zero. The bars differ in height, but the pattern is the same in each wave. Firms in newly connected municipalities are more productive than firms in places still waiting for service. The later 2016 to 2018 cohorts do not appear because no municipality was connected three years after them.

The intervals are wide for single cohorts, so I would not rank the waves. What matters is that the sign does not depend on one early group, and that is why I treat this comparison as the main evidence.

How does this relate to a standard regression? A pooled two-way estimate with firm and year effects gives 0.034 log points, about 3.4 percent, which sits at the low end of these bars. That is what a pooled coefficient does when arrival is staggered and differences grow with time: it mixes in comparisons against

already-served places. The table is in the appendix, together with the municipality-clustered standard errors.

What the cohort bars cannot show is whether the difference begins only when broadband arrives. For that, we need to open up the timing and look at the years on both sides of arrival.

Live cue: One bar per cohort; clean controls; all positive; pooled 0.034 at the low end; event time next.

Productivity rises before and after broadband arrives 2.4 min

Now open up the timing. The horizontal axis measures years relative to the arrival of commercial service in a firm's municipality. Zero is the arrival year, and negative values are the years before it. The year just before arrival is the reference point. The vertical axis shows the difference in log labor productivity relative to that year.

Before looking to the right, look to the left of zero. Several years before service, the coefficients are lower. They then climb steadily as arrival gets closer. The important feature is the slope, not the exact value of any point: productivity is already moving before the backbone becomes commercially available.

After zero, the coefficients continue to rise, and every post-arrival point lies above the pooled 0.034 from the two-way table. But the post-period begins on a trajectory that was already underway. In a staggered two-way event study the leads can also pick up effects from other periods, so I read the pre-period slope as a warning rather than a precise measure. The clean timing story would have shown the earlier and later cohorts moving together before arrival and then separating afterward. That is not the pattern here.

The rollout rule helps explain what we see. Municipalities with low current productivity but stronger projected growth received priority. If those projections captured real momentum, the early cohorts would be expected to catch up even without broadband. Their pre-arrival rise may be exactly the growth that placed them earlier in the queue.

The visible pretrend limits the estimates to a descriptive reading. We see a clear rise in productivity around the rollout, but this timing comparison cannot separate the broadband response from the catch-up path used to choose arrival dates. That is the interpretation-changing boundary in the design, and this is the point in the talk where it belongs.

There is still a useful pattern we can probe. If cheaper remote coordination is part of the pattern, the association should be larger for firms whose work relies more heavily on communication. The model turns that idea into a precise cross-firm prediction, and the final table asks whether the data follow it.

Live cue: Orient event time; pre-period trajectory; selection logic; descriptive boundary.

The model predicts larger gains for communication-intensive firms 2.5 min

The model puts one organizational choice at the center of the paper. Some tasks must stay inside, such as operating a machine. Accounting, software support, customer contact, and supplier coordination can potentially be bought from a remote specialist. Theta is the firm's share of tasks in that second group.

A specialist can perform those tasks better; that is the quality advantage phi. But using the specialist has two costs. The firm pays the provider's price, and it pays a coordination cost, kappa, whenever information and decisions have to travel across distance. Broadband reduces the coordination cost. It makes the relationship easier to manage without changing the specialist's underlying price or quality.

The inequality on the slide describes the interesting case. Before broadband, the quality gain is not enough to cover price plus coordination. The firm keeps the task inside. After broadband lowers kappa, the same outside provider becomes worthwhile, and the firm switches the eligible task to that specialist. *[pause]* The productivity gain comes from this change in who performs the task.

Now compare two firms that cross the threshold. One has only a few tasks that can move. The other relies heavily on communication and can apply the specialist's quality advantage to a much larger part of production. The second firm gains more. That is what the positive derivative says: within the switching region, the productivity gain rises with theta.

The threshold also gives us useful intuition about average effects. A firm that already used the specialist before broadband has no new sourcing change. A firm for which remote work remains too costly also does not switch. The response comes from firms near enough to the margin that lower coordination costs change the organization of production.

In the data, theta is the firm's 2008 wage-bill share in communication-intensive occupations. It is fixed before rollout and captures how much work could plausibly benefit from coordination at a distance. The model's empirical prediction is simple: the post-arrival productivity difference should be larger when that initial share is higher.

Live cue: Task choice; switching threshold; positive comparative static; empirical proxy.

The broadband association is larger for communication-intensive firms

2.4 min

This table takes the prediction directly to the data. Both columns use log labor productivity and the same firm and year effects as before. The first column allows the post-arrival difference to vary with the firm's 2008 communication intensity. The second gives the average post-arrival result as a benchmark.

Focus on the colored interaction row. The coefficient is 0.032. Because communication intensity is a share, a positive coefficient means that the productivity difference grows as we move toward firms with more communication-heavy work. The sign is exactly the one predicted by the task model.

The broadband coefficient above it, 0.020, is the fitted post-arrival difference for a hypothetical firm with zero communication intensity. For a firm elsewhere in the distribution, we add the interaction coefficient times its initial task share. So the economically interesting object is the slope across firms, not that intercept by itself.

A one-standard-deviation comparison puts the slope in useful units. The standard deviation of communication intensity is 0.20. Multiply that by 0.032, and the post-arrival difference is 0.0064 log points larger. In percentage terms, that is about 0.64 percentage points. Compared with the pooled 3.4 percent difference, it is roughly one-fifth as large.

Think again about the two firms in the model. Moving one standard deviation toward the communication-intensive firm does not replace the average result; it adds about 0.64 percentage points to it. This is a comparison that lives well inside the observed distribution, rather than an extreme contrast between a firm with no communication tasks and one made up entirely of them.

Fixing task intensity in 2008 is important for the interpretation. Firms are ranked by the work they did before broadband, so later workforce reorganization does not mechanically create the interaction. The initial task share delivers the cross-firm pattern the model told us to look for: the broadband difference is larger where communication-intensive work was initially more important.

Live cue: Orient columns; positive interaction; one-SD equals 0.64 points; model prediction.

Conclusion 1.2 min

Let me close with the paper's main message. Each of the first five rollout cohorts is more productive after broadband arrives than places connected later; the pooled summary, about 3.4 percent, is the low end. The difference is larger for firms that entered the rollout with more communication-intensive work: one standard deviation adds 0.0064 log points, about one-fifth of the pooled difference.

The model explains why that cross-firm pattern is economically meaningful. Better connectivity can make a distant specialist practical and move tasks across the boundary of the firm. Firms with more eligible tasks have more to gain when that coordination cost falls.

The broader contribution is to connect public infrastructure with the organization of production, not only access. That is why the productivity differences are largest where communication frictions bind. More generally, the value of a network depends on which production constraint it relaxes and which firms face that constraint most strongly. This broadens the policy question from coverage alone to the organizational frictions that determine whether firms can turn a connection into higher productivity.

Live cue: Every cohort higher; larger for communication-intensive firms; organization; broader implication.

Live cue layer

Frame title and first spoken sentence, in deck order.

1. **Key question** — Think about a small manufacturer that needs an accountant, software support, and help coordinating with suppliers.
2. **This paper** — Here is what we do.
3. **Roadmap** — I will make three stops.
4. **Broadband reached every municipality in eight waves** — The rollout gives us a simple piece of timing variation.
5. **Data** — We build the analysis from three administrative sources.
6. **The design follows firms as local backbone service arrives** — The design asks how a firm's productivity changes when local service arrives.
7. **Each 2011–2015 cohort is more productive after arrival** — This figure is the paper's cleanest timing comparison.
8. **Productivity rises before and after broadband arrives** — Now open up the timing.
9. **The model predicts larger gains for communication-intensive firms** — The model puts one organizational choice at the center of the paper.
10. **The broadband association is larger for communication-intensive firms** — This table takes the prediction directly to the data.
11. **Conclusion** — Let me close with the paper's main message.

Q&A appendix

conditional; excluded from talk time

The balanced panel describes surviving small firms

if asked

The source requires at least four consecutive observations, then reports a balanced panel of 9,400 firms observed in every year from 2008 through 2019. It does not explain that added balance restriction. The final panel accounts exactly for 112,800 firm-years, but it excludes entry, exit, and firms that fail the restrictions. Table 1 also reports an impossible employment combination: a mean of 23.4 and a maximum of 19. I do not use those employment moments in the talk. The productivity comparison therefore applies to selected survivors.

Projected growth helped determine rollout order

if asked

The 2010 Act committed the government to universal coverage by 2018. The RDDF then sequenced municipalities with a score that favored places below the median in current productivity but above the median in projected growth. Distance and terrain also constrained construction, but the paper does not isolate their independent role in timing. A natural next step would be to separate engineering variation from the growth score and ask whether productivity is stable before the resulting timing change.

The pooled two-way estimate sits at the low end

if asked

Table 2 reports 0.034 log points, and 0.030 to 0.033 with controls or regional trends. A pooled coefficient also compares against places already served, which pulls it down when differences grow. So the cohort comparisons carry the headline.

Rounded placebo entries cannot confirm the prose

if asked

Table 3 reports a placebo coefficient of -0.004 with a standard error of 0.002. Using the rounded entries, that is about two standard errors below zero, but rounding allows anything from about 1.4 to 3. The prose calls it statistically indistinguishable from zero, and the source supplies no exact p-value or confidence interval. I therefore do not read the placebo as a clean validation or rejection of the design.

Two numerical inconsistencies require correction

if asked

The abstract says 34 percent, the body says 3.4 percent, and Table 2 reports 0.034 log points. The region-trend prose also says 0.024 while Table 2 reports 0.030. The talk therefore uses the exact table units and cells: 0.034 log points for the baseline and 0.030 for region trends. Both narrative values require correction before publication.

Municipal assignment calls for municipal-level inference if asked

Treatment timing varies across 284 municipalities, while the baseline clusters by firm. Table 2 provides a municipality-clustered column: the point estimate remains 0.034 and the standard error rises from 0.002 to 0.003. Because arrival varies by municipality, I would emphasize the municipality-clustered uncertainty. It does not solve selection in the rollout score or the visible pretrend.

Larger gains require tasks to switch providers if asked

The proof covers the threshold-crossing case: remote sourcing is too expensive before broadband and becomes worthwhile afterward. Only then does lower coordination cost move eligible tasks to the specialized provider. If remote sourcing is worthwhile in both periods, or in neither, the sourcing decision does not change and this channel can generate zero gain. The proposition is therefore conditional.

The interaction is consistent with, not unique to, outsourcing if asked

The interaction shows that the post-arrival difference is larger for firms with higher pre-program communication intensity. That sign matches the model. But the data do not observe which tasks move outside the firm, payments to remote providers, or task reorganization around arrival. Other characteristics of communication-intensive firms could generate the same pattern. Direct sourcing measures before and after arrival would distinguish the proposed mechanism from those alternatives.

The six-year payback is not traceable from the reported inputs if asked

The paper compares a claimed annual flow of 410 million euros with construction cost of 2.3 billion euros. But it does not report the aggregate firm value-added input needed to reproduce the flow. The calculation also moves from a labor-productivity level association to TFP, annual growth, and fiscal return without causality, persistence, discounting, maintenance, coverage, or incidence assumptions. Gross output is not government revenue. I therefore exclude the payback and cost-effectiveness claims from the talk's conclusion.

Standing Q&A stance

Hear the whole question and count to three. Answer the strongest version in one sentence before adding detail. Concede a genuine limitation directly, then separate the paper's supported pattern from what remains to be shown.